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Elevating the game: Pathway structures and the operationalisation of talent identification and development systems across world netball

Fieke Rongen[a,b], Kevin Till[b], Megan Hill[b], Susan H. Backhouse[b], Dina C. Janse Van Rensburg[c,d], Jess Hayburn[b], Anita Navin[e] and Sarah Whitehead[b,f]

[a]Department of Sport Science, School of Science and Technology, Nottingham Trent University, Nottingham, UK; [b]Carnegie School of Sport, Leeds Beckett University, Leeds, UK; [c]Section Sports Medicine and SEMLI, Faculty of Health Sciences, University of Pretoria, Pretoria, South Africa; [d]Medical Advisory Panel, World Netball, Manchester, UK; [e]Coaching Advisory Panel, World Netball, Manchester, UK; [f]Leeds Rhinos Netball, Leeds, UK

ABSTRACT

Sporting pathways and talent identification and development systems (TIDS) are commonly used worldwide to identify and develop future elite athletes. However, limited research explores the structure of sporting pathways and the operationalisation of TIDS as part of them globally, especially in women's sports, including netball. Therefore, representatives from 10 global netball governing bodies participated in semi-structured interviews to help describe the structure of their netball pathway and operationalisation of their TIDS. Utilising reflexive thematic analysis three themes were generated: 'Netball pathways are multilayered but not always consistently connected', 'Traditional and forward-thinking talent identification and development practices', and 'Limited resources make talent identification and development challenging'. These findings provide organisations with an understanding of the structure of sporting pathways, and the operationalisation of TIDS in netball and the challenges of optimising them. To address these challenges, prioritising coach and practitioner education across the pathway to deliver inclusive, ethical, and appropriate practices to maximise player retention and player development is needed. Organisations are encouraged to consider how they can collaborate to maximise the use of resources within the sport.

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sport pathways; organisation; women sport; participation; resource

Introduction

The emphasis on identifying and developing young talented athletes in the pursuit of sporting excellence has led to national governing bodies and (professional) clubs investing considerable resources into Talent Identification and Development Systems (TIDS; Cobley et al., 2012; Rongen et al., 2018). Generally, TIDS involves five steps including, (1) detection, the discovery of potential performers who are not currently involved in the sport in question; (2) identification, recognising participants with the potential at an earlier age to progress to high development programmes; (3) development, a systematic combination of coaching, support, training and competition to progress athletes; (4) selection, the ongoing process of choosing individuals at various stages who demonstrate attributes suitable for future squads and teams; and (5) talent transfer, transfer from one sport to another sport where there are greater opportunities to succeed (MacNamara & Collins, 2015; Reilly et al., 2000; Williams et al., 2020). These five steps are often operationalised within everyday practice (i.e., identification or selection for the next step of a programme is influenced by performance in the previous development environment) and interwoven within a sports pathway.

A sport pathway refers to the overall sport structure, including recreational participation, TIDS, and elite performance, from childhood to adult retirement. A sport pathway outlines the stages of development, training and competition that allow athletes to experience, enjoy, develop and compete, typically covering a combination of education, community, and professional sport domains. Several development models (e.g., Developmental Model of Sports Participation [DMSP], Côté et al., 2007; the Foundations, Talent, Elite and Mastery [FTEM]; Gulbin et al., 2013) provide frameworks for what sport pathways can look like. These models show that most pathways commence with sports participation opportunities, focusing on playing, enjoying, and learning a sport, and continuing to adulthood for health and enjoyment benefits (Collins et al.,

CONTACT Fieke Rongen fieke.rongen02@ntu.ac.uk Department of Sport Science, School of Science and Technology, Nottingham Trent University, Clifton Campus, Clifton Lane, Erasmus Darwin, Room 285, Nottingham NG11 8NS, UK

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2012). However, in pursuit of athletes that can represent the sport at the highest level, many sport pathways employ a TIDS. The timing of talent identification to enter a TIDS can vary depending upon the sport, country and culture (e.g., 7 years in soccer, Noon et al., 2015; 15 years in rugby union, Till et al., 2020). TIDS are often structured pyramidally whereby at each stage of the system the number of places available decreases and the support provided within the programme increases (e.g., higher qualified coaches, increased competition). Furthermore, TIDS usually involve child and adolescent athletes, exposing them to intensified and specialised training and competition regimes, often requiring increased time (and sometimes cost) investment and continuous competition for limited places (Till & Baker, 2020).

Over the past few decades, academic reviews have presented potential problems associated with TIDS, including early selection, one-off identification practices, lack of clarity on 'what is talent' and selection biases (e.g., Bailey & Collins, 2013; Baker et al., 2018; Vaeyens et al., 2008). This research has questioned the effectiveness, healthiness and appropriateness of TIDS for youth athletes (Rongen et al., 2018), challenging current TIDS to adopt evidence-based practice (Pankhurst & Collins, 2013). However, as sporting pathways inherently have limitations in their resources (e.g., financial, personnel, facilities) (Till & Baker, 2020), TIDS are often necessary to ensure these limited resources are used in the most efficient way possible (i.e., targeted at 'the right' athletes). Although the academic reviews provide a broad description of TIDS, alongside the challenges associated with them, there is limited focus on how this research might be mobilised by those designing, implementing, and working on a day-to-day basis within TIDS. Specifically, limited research is available on how sport pathways are structured and how TIDS are a part of these pathways and operationalised in practice (Till & Baker, 2020). Such understanding would help inform sporting organisations of the best ways to design and deliver their sport pathways and TIDS.

A gender data gap exists in TIDS research with few studies including female athletes, and even fewer investigating exclusively female athlete cohorts (Curran et al., 2019; Hauser et al., 2024)). Gender differences play an important role in the development of young athletes and the findings from male cohorts cannot be directly applied to female cohorts (Curran et al., 2019; Emmonds et al., 2019). Despite a recent increase in research focusing on female athletes in popular sports, such as soccer (e.g., Andrew et al., 2024), the sport of netball remains notably absent from the evidence base. Yet, netball is one of the world's most popular women's sports with over 20 million participants worldwide (World Netball, 2024). Netball gained professional status in 2008 and now has professional or semi-professional leagues in Australia, New Zealand, the United Kingdom and South Africa. The increased professionalism has resulted in the development of player unions, enhanced sport pathways and TIDS to provide feeder teams for the professional systems (MacLean et al., 2022). As netball continues to professionalise and grow, increased investment is anticipated in the identification and development of the sport's future stars.

Research within applied sport science and medicine in netball has increased in recent years (Whitehead et al., 2021), but research considering talent identification (Jasni et al., 2022) and talent development (Nolte & Hollander, 2020; Robinson & Downward, 2009) is limited. Studies have identified differences in physical development (Simpson et al., 2019) and perceptual-cognitive and motor skills between levels of competition (Farrow, 2010). Additionally, player's perceptions of TIDS have been investigated in England (Robinson & Downward, 2009) and South Africa (Nolte & Hollander, 2020), with selection and specialisation occurring from 15 to 16 years of age (Robinson & Downward, 2009) and the lack of use of scientific support highlighted (Nolte & Hollander, 2020). Such studies give some insight into two nations, but little is still known about netball TIDS globally.

Therefore, the aim of this study was to describe how netball pathways are structured globally, with a particular focus on understanding how TIDS are operationalised as part of these pathways. Such understanding could inform the design and implementation of TIDS across world and national organisations whilst supporting practitioners in delivering inclusive, ethical, and appropriate practices to maximise player retention and development within the sport.

Methods

Research design

To describe the structure of netball pathways and operationalisation of TIDS within these pathways, a qualitative approach was taken within the Critical Realist (CR) paradigm. CR marries a realist ontology with an interpretivist epistemology (Easton, 2010), aiming to improve practice in 'the real world out there' (Pawson & Tilley, 2004). A qualitative approach suits CR's focus on understanding how something works, for whom, when and under what

conditions (Pawson & Tilley, 2004; Wiltshire, 2018) and aligns well with this studies' aim to understand how TIDS in netball work, with careful consideration for how this may be shaped within different nations with varying resources and contextual conditions.

Context of the study

World Netball (WN) is the international governing body for netball across five Regional Federations: Africa, the Americas, Asia, Europe and Oceania. This study was conducted in collaboration with WN following initial conversations with the Medical Advisory Panel and an interest was expressed to better understand the structure of sport pathways and operationalisation of TIDS across their national associations, with a view to support the development and provision of guidelines to support member nations. World Netball co-created the aim of the study and supported the recruitment process.

Sampling and participants

Following institutional ethical approval (application reference 111859), WN provided contact details for the national governing bodies (NGBs) of 20 nations across five continents. Participants were purposefully recruited via an email from the research team, with a WN representative included in this initial communication. To be eligible for inclusion, nations needed to have a recognised netball pathway (by WN) and be listed in the WN rankings. A deliberate attempt was made to recruit: (1) a representative sample of nations across the full WN rankings and (2) nations of varying geographical and socio-economic status. Within the NGBs, the request to participate in the study was directed to staff with knowledge of the pathway in their nation. Participants were provided with information detailing the study background, procedures, potential outcomes and intended use of data. They were assured of anonymity and confidentiality and provided written consent to participate.

Ten nations, with one participant per nation, agreed to participate. The final sample included five nations ranked in the top 10, four nations ranked 11–20 and one nation ranked 21–30 (WN Regional Federations: Africa $n = 1$, Americas $n = 3$, Asia $n = 1$, Europe $n = 3$, Oceania $n = 2$). In terms of income status (according to the World Bank, 2024), seven nations were classed as high income, two as upper middle income, and one as lower middle income. Eight participants were female and two were male. Participants' job roles ranged from chief executive officer ($n = 2$), (vice)-president ($n = 3$), head of performance/performance director ($n = 3$), coaching director ($n = 1$), and pathway manager ($n = 1$), and had an average of 4.4 ± 4.3 years (range 6 months to 15 years), experience in their current role. The two participants with less than 1-year experience in their current role had been involved in the organisation in different roles prior and as such were deemed to be sufficiently experienced to provide suitable insight. Six participants undertook their role on a voluntary basis, with two part-time and two full-time paid roles.

Data collection

An interview guide was developed after reviewing the relevant literature (e.g., Till & Baker, 2020; Cobley & Till, 2017; Ford et al., 2023; Nolte & Hollander, 2020; Relvas et al., 2010) and following recommendations for the structured-stage method of interviewing, consisted of three discrete sections (Rubin & Rubin, 2011). The 'warm-up' section aimed to build rapport, inviting the participants to share some general demographics and their journey and current role in netball. The second section focused on exploring (1) how netball was structured in their nation and (2) the player pathway and TIDS. Follow-up questions focused on when and how players typically entered the pathway, how players progressed through the pathway (including talent identification, selection and de-selection) and how the TIDS was resourced and structured. In the 'cool down' section, the researcher summarised participants' views as a check and challenge, and participants added anything additional. Throughout, probes and prompts were used to clarify responses, invite elaboration and elicit further detail, while keeping the conversation on topic (Smith & Sparkes, 2016). The interviews were conducted over Microsoft Teams (version 19.1.8). Online interviews are deemed appropriate and particularly suitable when participants are geographically dispersed (Lobe et al., 2022). Interviews lasted between 42 and 90 min (58.8 ± 14.4 min) and were audio-recorded for subsequent verbatim transcription, providing 184 pages of single-spaced text for analysis.

Data analysis

Within CR the researcher is positioned as actively making sense of the data in a recursive and iterative process, aligned to Braun and Clarke's (2019) description of reflexive thematic analysis, wherein researchers work to generate themes (patterns of shared meaning) to answer the research question. The analysis employed an abductive approach (Lusted, 2018), acknowledging our extensive familiarity with existing literature which informed the study design and data analysis, while

simultaneously remaining open to unexpected and novel ideas regarding the structure of the sporting pathways and operationalisation of TIDS within netball. Data analysis started with the first and third author familiarising themselves with the interview transcripts through reading through them multiple times, noting down any reflections in a separate document to help guide later discussions. In line with the purpose of the study, the third author then coded the transcripts semantically, focusing on what participants explicitly expressed regarding their pathway and TIDS practices with a smaller number of latent codes, focusing on more implicit or conceptual meaning (Braun & Clarke, 2021).

Following this, the first author reviewed the codes against the transcripts and the first and third author then met to discuss the codes and generate themes and subthemes by grouping codes around central organising concepts. The full research team reviewed the thematic structure (i.e., themes and subthemes) by going back-and-forth to the data, allowing for reflection to further make sense of the findings, and group discussion then led to the themes and subthemes being reconfigured a number of times (Braun & Clarke, 2019). For example, we initially started with topic summaries offering descriptions of what had been said in relation to the pathways and TIDS practices. Through further analytical work, drawing on our knowledge of the existing literature, final themes and subthemes were developed which present shared meaning about how nations globally made TIDS work. The first author then took the lead in the write-up process, which allowed the team to collaboratively and iteratively define and name the themes and subthemes. In addition, based on the content of the interviews, a schematic of the pathway of each nation was drawn to depict the different levels (both recreational and performance) and age-groups. Participants were given the opportunity to review the schematics for accuracy, with some minor amendments requested.

Markers of quality

We followed Ronkainen and Wiltshire's (2021) recommendation and draw on Maxwell (2012, 2017) typology of validity to ensure the quality (i.e., empirical adequacy), plausibility and utility of our findings. Regarding adequacy, empirical validity (i.e., how factually accurate is the qualitative research account) was ensured through clearly describing the research process in the methods section as well as through recording and precise transcription of all interviews, with transcripts thoroughly checked for accuracy against recordings. To ensure interpretive validity (i.e., the degree to which the presented narrative accurately reflects the perceptions and experiences of participants), we foregrounded participants voices and included quotations throughout the results to show the interpretations are empirically grounded. In addition, pathway schematics were shared for feedback. In terms of plausibility, to ensure theoretical validity (i.e., do we provide a coherent account of the events in the real-world and participants experiences of them), the research team engaged in frequent peer-debriefing and critical discussions to review and challenge initial interpretations (Ronkainen & Wiltshire, 2021; Smith & McGannon, 2018). During these discussions, the research team aimed to position and interpret findings considering existing literature, further increasing reflexivity (Ronkainen & Wiltshire, 2021). Finally, in terms of utility the reader should assess whether the findings make sense to them and could be used to guide action in practice. In addition, in the discussion we provide recommendations and interpretations to guide practical actions in the real world.

Findings

Three main themes, each with three or four subthemes, were generated. The first theme outlines how 'Netball pathways are multilayered but not always consistently connected', with four subthemes: (1) educational pathway: netball for all, (2) club pathway: netball for most, (3) representative pathway: netball for the chosen few and (4) lack of connection and coherence between domains. The second theme identified 'Traditional and forward-thinking talent identification and development practices', with three subthemes: (1) talent identification: selectors identify by 'seeing players' in action, (2) aiming for late selection into an open-fluid system and (3) selected: greater investment expectations and an aspiration for holistic support. The third theme explored how 'Limited resources make talent identification and development challenging', with three subthemes: (1) a lack of human resource to effectively run the system, (2) physical resources are limited and (3) lack of competitive structure (Table 1).

"Netball pathways are multilayered but not always consistently connected"

All participating nations have a netball pathway that includes participation, talent identification and development, and high-performance adult elite netball. Figure 1 presents a schematic of the pathway for each nation. Whilst elements of pathways are similar, there is variation concerning the timing and setting of the introduction to netball, and the timing and volume of talent identification and development

Table 1. Summary of themes and sub-themes for the structure and operationalization of netball pathways.

Theme	Sub-Theme
Netball pathways are multilayered but not always consistently connected	• Education pathway: netball for all! • Club Pathway: Netball for most • Representative Pathway: Netball for the chosen few • Lack of connection and coherence between domains
Traditional and forward-thinking talent identification and development practices	• Talent identification: Selectors identify by 'seeing players' in action • Aiming for late selection into an open-fluid system • Selected: Greater investment expectations and an aspiration for holistic support
Limited resources make talent identification and development challenging	• A lack of human resource to effectively run the system • Physical resources are limited • Lack of competitive structure

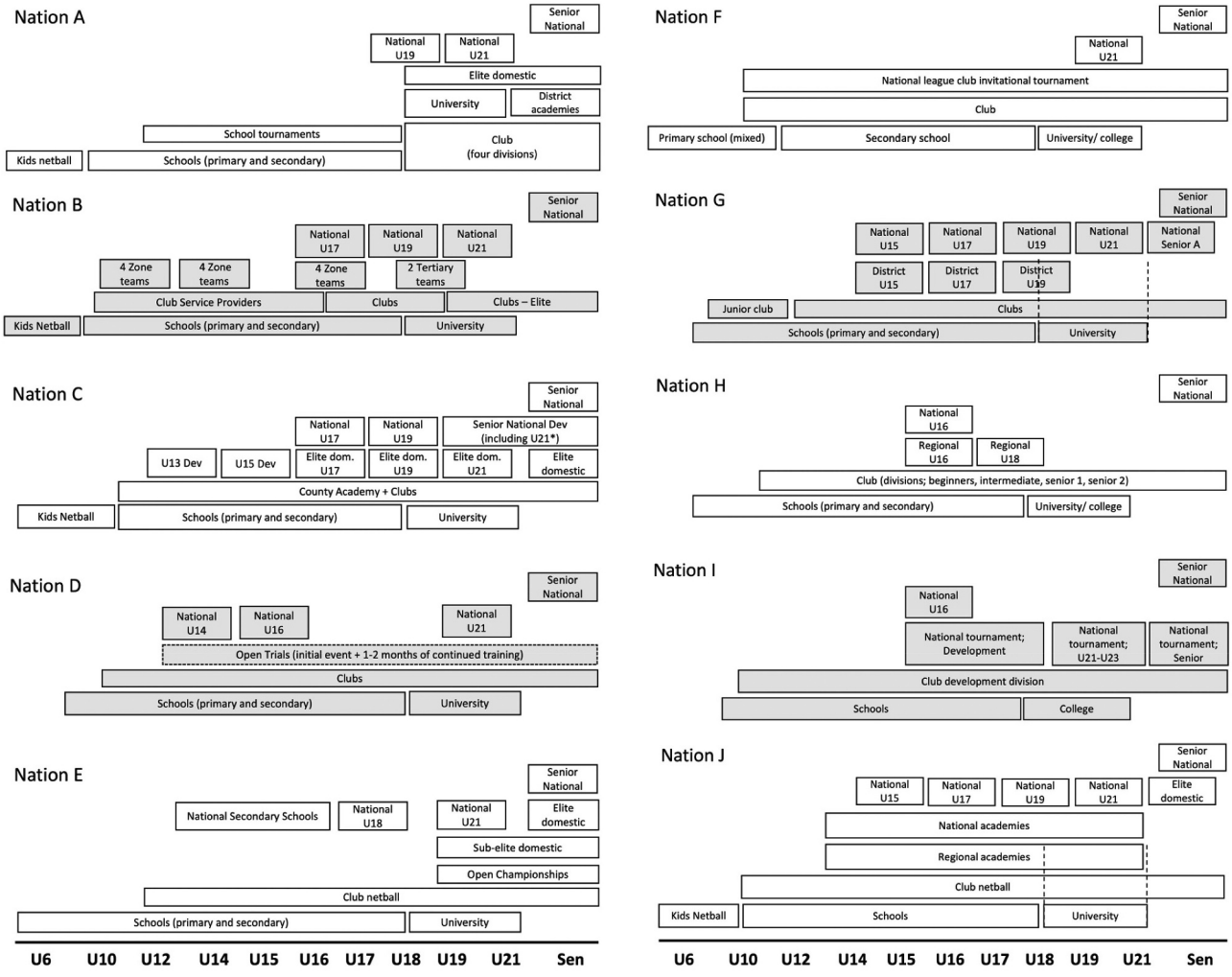


Figure 1. A schematic of the pathway for each nation. The schematic indicates the different components of the recreational and performance pathways, by age (horizontal) and level (vertical). Movement occurs through the vertical component with simultaneous involvement of athletes across different levels. U6 = under 6 years old, U10 = Under 10 years old, etc. Sen = senior. Dev = development. Dashed line indicates nations where university netball was reported to be part of the performance pathway.

opportunities alongside adult and high-performance netball. Globally, pathways for netball are positioned in three main domains: (1) education (school, higher education), (2) clubs and (3) representative teams. Each of the domains fulfil several core functions in terms of collectively supporting the development of netball players across all levels. Yet, at the individual level, for many nations, players are simultaneously participating in multiple domains which are not cohesive or well connected, creating challenges for the players navigating their journey.

Education pathway: netball for all!

Educational contexts were highlighted as an inclusive entry point and a key setting providing participation and development opportunities *to all*. In all participating nations, primary schools introduced children to netball through modified versions of the sport; '*netball is and has been for some time on the school's curriculum*' (Nation H). They also all offered opportunities to play netball within secondary school and through further/higher education providers (e.g., college, university); '*the schools' netball caters from ages 5 until 18 and then we have universities that also are associated members. They cater from 18 until 21/23*' (Nation A). Competitions or festivals were available within primary and secondary schools across nations. For some this was non-competitive; '*they have sort of competition, non-scoring, non-competitive competition from age of about 10 going on 11*' (Nation B), whilst others had small internal or provincial competitions, and even national competitions; '*most of the schools will have their internal competitions . . . but there is the national school games competition*' (Nation I).

From secondary school through to higher education, the purpose of the netball offered was twofold; play recreationally or talent development for performance netball. Three nations specifically highlighted that higher education provision was part of their performance pathway, whereby higher performing players enrolled in universities to continue playing. Several nations mentioned a wish to integrate the university and high-performance netball programme, but only one nation had done so; '*they have to play university as well as professional and they cannot decide to play one or the other*' (Nation A).

Club pathway: netball for most

All participating nations had club netball, but like schools, club netball differed across the nations. Club netball offered opportunities for those who want and are able to, continue playing netball at various levels (e.g., recreational, talent, performance) into adulthood. In eight nations, entry to club netball started between the ages of 8–11 years. However, in two nations, club netball started from the age of 15 years (Figure 1). Although some players may be first introduced to netball through a club, club netball was predominantly seen as a place to build on school involvement; '*a lot of the coaches coach in the schools and these coaches, many of them are affiliated to certain clubs so they then bring these players in from the schools*' (Nation B). In some nations, children were specifically recruited to clubs based on their performance in school netball; '*so once the clubs see you playing at the secondary level they might invite you to play for their club*' (Nation D). Typically, club netball was seen *as additional* to school netball. For example, in all participating nations that had clubs for juniors, it was clear that children play netball in both their school and their club; '*play at secondary school as well as they play for their club*' (Nation F).

Representative pathway: netball for the chosen few

All participating nations had some degree of youth district, regional or national representation that forms part of their pathway (see Figure 1). Two nations had a progressive national programme (e.g., U15, 17, 19, 21 senior), whereas others had national representation in either U16 or U21 age categories. In all participating nations, representative netball was positioned as netball for *the chosen few*.

Lack of connection and coherence between domains

At the systems level, the multiple domains collectively support the development of a body of netball players (participation and performance). Players often simultaneously participated in multiple domains, yet, in most nations, the multiple domains (education, club, representative) are not cohesive or well connected. From a player experience and development perspective, this poses challenges around logistics, training and competition load, and the aims of each domain. For example, players selected for a representative team may be given a specific training programme by the representative coach, but they '*have to rely on the club coaches to help deliver this*' (Nation A). Equally, training and game demands increase, beyond desirable levels, for players who participate in multiple layers of the pathway; '*still be playing at school, they'd be playing at club, and if they're involved in a [elite domestic] team then they would be attending a weekly session for that*' (Nation C).

Additionally, when players are selected to represent a team, schedules often clash:

> When you're at national training you basically can't get to your club practice so if you're preparing for a tournament some players would make a request to the coach if I can have an early out so I can go to my club practice because we have a game coming up. . .so they would leave the national practice and go over to the club practice so they're kind of doubling up. *(Nation D)*

"Traditional and forward-thinking talent identification and development practices"

Talent identification and development practices in netball are traditional and forward-thinking. From a talent identification perspective, there is a reliance on traditional means of selectors 'seeing players' and judging their potential (often using limited and non-holistic criteria). At the same time, the majority of nations strive for an open/fluid system where selection decisions are made at a later age and are not necessarily final. In terms of talent development, once selected, time invested in training and competing increases, but early specialisation was discouraged and a strong aspiration to offer holistic support was espoused but often not delivered.

Talent identification: selectors identify by 'seeing players' in action

All participating nations had some form of talent identification and selection, although there was variation in terms of where, when (i.e., what age), and for what purpose. Most nations focused on identifying and selecting players for long-term development, yet some nations explained they did not have a continuous identification process; rather, selecting players at key timepoints, such as prior to a tournament. Identifying and selecting players took place in multiple ways including training and competition observations, invited and open trials, camps and player screening. Although selection varied by nation and even by region within nations, '*screening is run different in every different county*' (Nation C).

All participating nations used a process of selectors '*seeing a player*' and judging their performance and potential. Half of the nations explained they have a selection committee or panel; '*director's selections, she has a group of selectors*' (Nation A) whereas the other half relied heavily on the judgement of one coach/scout. In either case, the selector(s) were usually '*experienced coaches*' as '*they know what to look for in a good player*' (Nation D). In terms of what selectors looked for when identifying players, only half of the nations mentioned having a clear set of criteria and a selection policy document. Others expressed not having clear formalised criteria or that criteria changed depending upon the aim of selection; '*it just depends on the year and what we decide to do*' (Nation D). For those with criteria, most focused around technical, physical and tactical performance in drills and games. For a few nations, the criteria encompassed a more holistic view of the players, including psychosocial characteristics:

> We have a set of criteria that we look for all of the player's and it covers things around, the basic netball skills, movement skills, lots of things around team working, resilience, their ability to cooperate with coaches and other people in the environment, physical characteristics, and injury. *(Nation J)*

In addition, some nations shared how their clear criteria were not only intended to aid talent identification but also support coaches in player development practices:

> We created a benchmarking process . . . we looked at some of the key criteria, some movement based ones, some game sense based ones, some attacking and defending ones, and some positional specific, and we have put some criteria. . . so it was 1, to help with selections and scouting and 2, to help coaches understand, okay these are some of the things I need to be working on with my players. *(Nation G)*

To prepare their selectors, five nations had a certified course which selectors attended to increase knowledge, and help ensure selected players meet the expected criteria; '*ideally, the selectors are people who've all attended a player identifier course because that would give them kind of a background as to the criteria we're assessing the players against*' (Nation C). Other nations referred to a system whereby, coaches were 'sent information' or had opportunities to discuss the processes; '*we do have guidance and we present within a high-performance forum every year . . . around what we are looking for, and we communicate that regularly*' (Nation E).

Aiming for late selection into an open-fluid system

Many of the participating nations tried to deploy late (i.e., mid-late adolescence) selection, apply an open/fluid system in terms of multiple and recurring opportunities to be (re)selected and allowed players to self-nominate. Although nations differed in the ages of identification/selection, there was agreement that this started from secondary school: '*we say unless you're 13 we shouldn't really be putting you in a performance environment yet*'

(Nation J). Further, there was a sense that the system should remain fairly open and ideally more final decisions should not be made until the U18s or U21s levels.

> We would really start at the national secondary schools level but that would to me be a very blurred entry if that makes sense like I really don't think that you can have any assurance until you're probably at under-18 or under-21 level as to who's gonna really make it through. *(Nation E)*

Second, many nations expressed that they were able to operate or were '*hoping to be a system where there's multiple entry and exit points*' (Nation E) where '*it's always an open doorway*' (Nation C) and where players '*could come in at any point if you're at the right level, it's intended to be quite fluid*' (Nation J). This was further reflected in very few nations describing deselection processes from the TIDS. Although it appeared possible to release players, for example for behavioural or performance reasons, it was rare that this occurred and more common that players would deselect themselves due to personal commitments or because they had not advanced to the squad they hoped for

> There would be others who are removed or deselected on behavioural traits probably you know just they're not really in the mindset or won't have the mindset to progress any further so I think often players in the end, the deselection for the most part takes care of itself because a player will reach a ceiling of what they're capable of, or they're prepared to do. *(Nation E)*

Rather than deselection, in many nations, players were dropped into a lower level of the pathway to continue to play and work on their skills with the opportunity to retrial at a later stage for the more elite level.

> There is a conversation between the coach and the manager, and they would let the administration know these persons were deselected and they have the conversation with them as to why they need to go back and work on and that kind of thing to come back. *(Nation H)*

In line with this, and to support a more fluid system, a small number of nations allowed players to self-nominate for inclusion in talent identification and selection trials.

Selected: greater investment expectations and an aspiration for holistic support

When players were identified/selected, they were expected to invest more time due to the increase in training and competition volume and intensity. As players progress, they could be training and competing for their school, club, region, club and national team:

> If you're 13/14 year old and you're playing a bit more netball, you'd probably have one session with your club, you might have your school practices, you'd probably have a game at the weekend . . . if you then entered into district, you're probably looking at an additional evening a week, so you probably looking at an evening of district, an evening of club, maybe a school practice after school maybe if you're district and club maybe one or two matches at the weekend. If you got through and let's say you're an 18 year old and you're playing for [Nation] under-19's and you're going to university etcetera, it does ramp up quite quickly. *(Nation G)*

As a result, for most nations managing training volume and activity was challenging due to the numerous training and competition contexts a player could participate in.

At the older ages, or the more elite level, specialisation in netball was expected, as a way for players to balance their sport and other life commitments; '*around the 16/17/18, that's when they're maybe going to start streamlining and partly also because of balancing exams, balancing life*' (Nation G). Yet, at younger ages, all of the participating nations explained that they would not expect, and discouraged, specialisation solely in netball as this was not judged to be in the best interest of the player. Despite these efforts, and because of the increased demands once part of a TIDS, many suggested it was difficult for players not to specialise, even at the younger ages; '*it's just so difficult to keep other sports unless it's a summer sport*' (Nation C).

All participating nations explained that TIDS players were offered additional support. There was a strong aspiration to provide a holistic development programme, including '*technical work, tactical work, video analysis, nutrition, injury prevention, performance lifestyle, all of that stuff, sports psychology*' (Nation J). However, the capability to deliver a holistic development programme varied greatly between nations. For the majority, holistic support could only be offered to the most senior elite team (National) or more occasionally at camps: '*you'll have a little bit of input at the camps, we don't have enough resource for that to then spread further from that*' (Nation G). For some, particularly those with lower economic status, even accessing support at the highest level was a challenge due to the resources available.

"Limited resources make talent identification and development practices challenging/precarious"

To understand how talent identification and development was operationalised, it is important to

acknowledge the context in which netball talent development was expected to operate. All participating nations expressed an ambition to create a fluid, inclusive and effective system, which offered higher quality training and competition with increased support. However, all participating nations, regardless of economic status, highlighted a lack of funding, impacting resources available. One nation shared how they were '*trying to get some level of sponsorship into the sport*' (Nation H) to provide an income stream, whereas two nations explained funding is so limited that they have to fundraise for their sport: '*it's so difficult trying to raise money for the million things what we have to do*' (Nation D). A lack of financial resources resulted in a lack of human and physical resources. Below we will discuss the main resource challenges and demonstrate how they permeate all aspects of the TIDS, and pose persistent challenges in effectively identifying, selecting and developing talented players. Although this applied to all participating nations, these challenges were exacerbated for those countries with lower economic status.

A lack of human resource to effectively run the system

A lack of human resource to effectively run the system was identified as the biggest challenge by the participating nations. Specifically, there was a lack of (1) a professional workforce, (2) training and knowledge within the existing workforce, and, particularly in the smaller nations with a lower economic status, (3) depth in the player talent pool.

Due to limited resources, even at the elite level, most participating nations were unable to pay or even subsidise the costs for their players; '*players do not get paid, it is all voluntary*' (Nation B). In some circumstances, nations had '*provision for hardship funds*' (Nation G) to try and support players from poor socioeconomic backgrounds. However, as players would typically have to work alongside their sporting commitments, this impacted the nations' ability to develop talent: *'as soon as they start with jobs they tend to go and play socially because they do not have the time to properly train*' (Nation A). The lack of financial support for players was also seen to affect the depth of the talent pool. Several nations recognised that players who could not afford netball, or had to work, would not be seen, therefore, not identified or selected; '*if you don't have the funds to attend a tournament you will not be seen unfortunately so it's mostly the players that have the funding at the end of the day you know progress*' (Nation A).

Equally, for most nations, coaching positions within TIDS were largely voluntary, with some coaches receiving honorary payments: '*probably 80% are voluntary because we only pay them when we're on tour or at a training camp*' (Nation F), or coach development opportunities. Only a small number of nations were able to employ senior coaches; '*there will be a standalone kind of head coach looking after the international team, and then our pathway coaches under-21 and down they get a remuneration, but it's far more voluntary than anything else*' (Nation G). In most participating nations, netball coaches were former players of the game, possessing a good understanding of the sport. However, in most of these nations, coaches were not qualified, although there was a push to start making coaching more professional: '*we have a lot of persons who would've played netball and would just start coaching, what we're trying to do now is to train those persons to be qualified coaches*' (Nation D). Yet, as coaches were also almost always volunteers, upskilling and educating coaches was challenging: '*They will attend the workshops but that's where it will stop*' (Nation A).

Similarly, the provision of holistic support services, for many nations, relied upon volunteers, particularly in the countries of lower economic status: '*what we tend to do is ask persons with any qualifications to volunteer at strength and conditioning, nutrition, massage therapy and physiotherapy*' (Nation H). Others worked with partners (e.g., universities) to outsource the delivery of this support in the format of workshops. Beyond the development of their players, some nations also expressed how a lack of selector workforce stopped them from keeping their pathway as open as they would like: '*what we could do with having is more workforce so that in between that trial and the next one, we can see some of these players in action*' (Nation G).

Physical resources are limited

Some nations highlighted a lack of physical resources, with four nations explaining they had problems accessing (quality) facilities, even for their top players: '*yes they train in a central location, unfortunately it's still on concrete so it's very damaging*' (Nation D), although this was more evident in nations of lower economic status. The limited training facilities in combination with the size of some nations, meant many nations described increased travel demands on their players as another challenge. Players and coaches had to travel considerable distances to attend training and competition; '*in the rural areas you might drive 1 to 2 hours to go to games*' *(Nation A)* and '*we have players who could be 50 miles away having to travel for their session so it's tough*'

(Nation C). There appeared to be no policies in place to prevent players from extended travel times, but some nations were trying to mitigate travel time where possible.

> We try to make it work that they have two or three games over a weekend, and they might travel, so in terms of their district competition, that's why we did a North, Central and South, so that they weren't having to go from the [region] down to [other regions] or whatever that looks like. *(Nation G)*

Lack of competitive structure

A final challenge in many nations, also driven by a lack of financial resource, was the absence of a professional league, or other appropriately high-level competitive structure. Without such a structure it was perceived to be difficult to prepare players for 'what is next':

> In the last couple of years we have created a lot of, some competitions to provide for the fact that if you're a [National] player and you're not at [next level up] level, what do you do to actually bridge the gap? Either to prepare for international events or to prepare and compete in a way that then will maybe get you a contract in the future or just kind of raise your standard. *(Nation G)*

One nation explained that when identifying players at the highest level, they search for players from their nation overseas due to the lack of depth playing in their own league:

> When we are looking for our talent we tend to look overseas because it's a lot more, and they are at that high performance space, they are playing at that high level netball competition. With all due respect to ours it's just that we don't have that calibre, that standard of play in our domestic league … unfortunately our players are all from overseas cause we don't have the depth here. *(Nation F)*

Discussion

To support sport organisations in designing and implementing evidence-based sport pathways, an understanding of their structure as well as the operationalisation of TIDS as part of them is required. This study aimed to describe how netball pathways are structured globally, with a particular focus on understanding how TIDS are operationalised as part of these pathways. Interviews with 10 nations generated three themes: (1) 'Netball pathways are multilayered but not always consistently connected', (2) 'Traditional and forward-thinking talent identification and development practices' and (3) 'Limited resources make talent identification and development challenging'. These findings highlight that netball pathways globally share common features but vary in terms of their pathway and TIDS structure and operationalisation (usually due to the resources available and/or nation size/geographical location). A lack of resources was highlighted as a universal challenge that impacted nations' ability to optimise player identification and development practices. This study offers an insight into the strengths and limitations of netball pathways and TIDS, which in turn may help world and national netball organisations develop policies towards better informed pathways and support practitioners to deliver inclusive, ethical and appropriate practices to maximise player participation, development and performance within the sport.

In all participating nations, the netball pathways demonstrated common features with popular athlete development frameworks (e.g., DMSP, Côté et al., 2007; FTEM; Gulbin et al., 2013). Similar to existing models, the introduction and participation in netball commenced in schools and clubs and focused on having fun and effectively learning skills through modified versions of the sport (Miller et al., 2017; Spencer & Wolf, 2017). In line with the DMSP and FTEM, players are provided opportunities to continue playing netball into adolescence (post-11 years) in secondary schools and clubs, either for recreational reasons or to start investing in talent development or performance. This approach recognises that young netball players have varied motives for participation (e.g., wellbeing, personal referenced excellence, elite referenced excellence; Collins et al., 2012). Although netball pathways seemed focused on transitions to performance, it is important that national governing bodies recognise these varied motives as many sports, including netball, demonstrate a decline in girls participation during adolescence (Emmonds et al., 2024; Litchfield & Elliott, 2020). Given the prominence of netball within schools highlighted in this study (Figure 1), and in a previous report on talent development in English netball (Robinson & Downward, 2009), this may be a key context for retaining netball players.

All participating nations had a form of TIDS, positioned as netball for the chosen few, following the pyramidal structure common in many sports (Till & Baker, 2020). However, variability in the TIDS was apparent between nations, particularly for smaller sized nations (with a smaller number of available players) and those of lower economic status. For example, two nations consistently provided national representative opportunities across age groups, whereas others only had one national age-group representation in their TIDS (e.g., U16; Figure 1). Participating nations with national representation across three or more age-groups were all 'high-

income' status countries, whilst those with fewer age-groups were of a lower economic level. As resources largely determined age-group provision, nations may want to consider where best to invest to support transition to the senior level (e.g., is national U16 representation the most appropriate age group for a low-resourced nation?). A novel finding was the role of higher education in nations' pathways, providing an opportunity for netball players to combine sport and education, through either participation or performance pathways. Higher education performance pathways may mirror the US College system and differ from other popular team sports (e.g., soccer) TIDS, which may be a consideration for netball TIDS given the resource challenges reported.

Overall, a common principle to consider for the represented nations was the connection and coherence between the pathway domains (i.e., education, club, representation). Our findings highlighted that some netball players participated in multiple domains simultaneously, potentially creating logistical challenges (e.g., travel, clash of training), inconsistencies in the aims of training programmes and high training and competition loads. Such factors may impede player development and even affect netball participation due to factors, such as injury (Emmonds et al., 2024). Similar challenges, described as 'organised chaos', have been reported for school sport student-athletes (Scantlebury et al., 2021) and academy rugby union players (Phibbs et al., 2018), and may have implications for player health, development and performance (Bergeron et al., 2024). Collaboration across domains has been recommended in other sports (Mathorne et al., 2021), which could be applied in netball but is challenging, especially in under-resourced nations.

Focusing on how talent was identified and selected, the 'traditional' practices used (e.g., coach/selector observations) for national representation, without a clear selection criteria, suggest that talent selection (i.e., current performance) processes are the most prominent. Research (e.g., Baker et al., 2018) has questioned such approaches, recommending to move away from 'traditional' performance-based selections. Two considerations have been proposed including (1) the understanding of talent as a dynamic, emergent, multidisciplinary and non-linear trait (Baker et al., 2018) and (2) the timing of identification/selection decisions based on the time length between youth and elite-level performance, whereby the sport (e.g., rules, demands) may change. In-line with current literature that compares between levels of competition (e.g., Farrow, 2010; Simpson et al., 2019), focus is upon the technical, tactical and physical development characteristics, with few nations mentioning psychosocial characteristics within their identification and selection practice, which are deemed paramount for future success (Dohme et al., 2019). Furthermore, by focusing on subjective coach opinions may bias selection decisions, especially among inexperienced and uneducated coaches (e.g., Xiang et al., 2024). Therefore, organisations should establish holistic criteria for talent identification using signs (player characteristics), samples (netball performance) and subjective expert opinion for netball player development that acts as both a talent identification and player development tool (Barraclough et al., 2022; Till & Baker, 2020). Notably, none of the nations discussed talent detection nor talent transfer processes as part of their TIDS. Whilst likely requiring additional resource, netball NGBs may benefit from considering whether these approaches could offer worthwhile ways to widen their talent pool.

Having identified these issues, it is important to recognise the existence of a number of forward-thinking practices for talent identification described by the participating nations. These practices included delayed (post-adolescence) identification/selection, coach- and self-nominations and using a system that allows multiple entry routes may overcome some of the challenges associated with talent decision-making. Interestingly, although some of these practices are at least partly driven and/or necessitated by a lack of available resources, they align closely to previous research recommendations around optimising TIDS (e.g., Baker et al., 2018; Till & Baker, 2020) and have so far not often been documented as being applied in real-world systems.

Talent development aims to provide athletes with a suitable learning environment to realise their potential (Williams et al., 2020). All participating nations aimed, through their TIDS, to deliver higher quality training and competition with increased support (e.g., coaches, holistic player services). However, practices were highly variable and constrained by several common and consistent challenges that hindered effective implementation and delivery. First, delivering high-quality training opportunities is often costly, requiring the support of facilities, personnel, and resources, which were often unavailable. Second, many nations relied on voluntary (or small honorary payment) coaches with limited coaching qualifications. Third, whilst holistic player services were often considered important, including strength and conditioning, physiotherapy, nutrition and psychology, they were mostly delivered via volunteers or not provided at all. Previous research in South Africa (Nolte & Hollander, 2020) supports these two elements as only ~50% of players reported their coaches to have adequate knowledge of netball for competition performance. In

addition, these same players made limited use of the scientific support provided outside of physiotherapists and medical doctors (Nolte & Hollander, 2020), perhaps indicating the need for education to players on the importance of such services. Finally, as is often the case with TIDS, the ability to deliver high-quality training experiences in central venues presented a challenge in most nations, requiring players and families to incur additional time and costs.

Practical implications

Effective TIDS require a broad base of a system, and it is imperative that nations establish and encourage early engagement in the sport to enable long-term retention. As such, delivering enjoyable, developmental programmes in a safe and caring environment is important (Till et al., 2022) and schools could be an ideal context to deliver such programmes. From a talent perspective, a balancing act is needed between establishing a base of the pathway and appropriate TIDS given the resources available. Nations should consider delaying selection opportunities to post-adolescence, to help reduce the pressures placed on young players, who often train and compete across multiple environments, resulting in high, unmanageable workloads. The use of an open and fluid system that uses a self-nomination process is a further recommendation. Furthermore, nations could consider the use of higher education (i.e., universities) as an important part of the TIDS to support transitions to high-performance. To support talent identification/selection decision-making, establishing multidisciplinary criteria for use across multiple stakeholders as an identification and development tool would be beneficial (Till & Baker, 2020). Coach, selector and support staff education and development would be a key strategy for enhancing the existing and new workforce to improve talent identification and player development opportunities. To support effective coach education and development, the make-up of the workforce (i.e., volunteer dominant) should be considered; future research should consider the experiences and perceived needs of the workforce, as well as realistic ways of engaging with them. Given the prominence of the school context in netball pathways, it would seem pertinent to include physical education teachers and those delivering in schools in such endeavours too. A greater level of communication and collaboration between the various contexts, and ultimately coaches, that youth athletes can simultaneously be involved with would allow for a more considered management of workload as well as synergy of developmental experiences on offer.

Achieving these suggestions against limited resources (i.e., financial, human, physical) reported by nations, particularly those of lower economic status, is challenging. Limited research is available that explores resource use within talent identification and development. One potential solution that emerged from the findings is that netball organisations should explore opportunities for collaboration with other organisations (e.g., universities) to develop and deliver talent development programmes towards the performance domain and to provide holistic sport science support for athletes that may be combined with a dual-career approach. Such collaborations would also support applied research, reducing the research-practice divide and ensuring research impact (Jones et al., 2019). All these recommendations could be considered by WN as the world governing body to support nations' pathways and TIDS. The development of frameworks and criteria to support professional coaching, coach education, pathway development, and talent identification and development systems, policies and practices would enhance their sport globally.

Limitations

Whilst this is the first study to explore netball pathways and how TIDS are operationalised as part of these globally, limitations exist. Firstly, only one representative from each nation was interviewed, and although they understood the pathway, they were most likely not involved in delivering all aspects of the pathway. In addition, two of the participants had less than 1 year in their current role; therefore, despite being involved in the organisations in different capacities prior, they may have developed more of an understanding of the pathway with more time in the role. Although not the purpose of the current study, future research could explore a single nations' pathway and TIDS in greater detail by focusing on how the various stakeholders involved deliver it and how they work together. Equally, this study has not provided an insight into what it is like for netball players to navigate the pathway and TIDS, which would be a valuable addition to the knowledge base and a worthy topic of further study. Second, although nations were approached by the research team, they were aware the project was in collaboration with WN, and – despite assurances of anonymity and confidentiality – there may have been some power at play to not share all elements equally (i.e., not be overly negative). Additionally, only 50% of the nations contacted agreed to participate, limiting the sample and therefore representation of different TIDS

within the sport. However, the final sample was representative of different rankings and income status. Finally, collectively describing the structure of the netball pathway and operationalisation of the TIDS in 10 countries (whilst also a strength) is difficult due to the contextual factors that may impact pathways and TIDS within the respective nations. Yet, 'generalisable laws' are not the purpose of CR-oriented research. Instead, we aimed to show sensitivity to contextual differences, through depicting individual pathways and presenting nuances in our results where possible, whilst also offering insights into collectively shared features and challenges to invite others to draw parallels to their own pathways and TIDS and explore whether there are any 'lessons learned' (transferable generalisability, Smith, 2018). Here, it is important to note that although we aimed to explore how things worked differently across different contexts, there were some clear universal findings which we have focused on more strongly in our discussion. For example, it was evident that all nations, regardless of economic status, were impacted by limited resources in a variety of ways. Further investigation into the impact of economic status within the pathway and the distribution of resources is warranted.

Conclusion

Understanding the structure and operationalisation of netball pathways and TIDS is important to support the increasing popularity and professionalism of the sport. Findings demonstrated the multilayered nature of netball pathways, which are not always consistently connected, alongside both traditional and forward-thinking talent identification and development practices, though limited resources make talent identification and development challenging. Netball commences in schools and progresses within secondary schools, higher education, and clubs providing both recreational and developmental opportunities leading to representative representation for the chosen few. Yet, domains often lack connection, potentially limiting player development. Talent selection (rather than identification) appears common, whereby representative selections were made on current performance via coaches/selection panels' observations with limited established criteria to support decision-making. However, TIDS generally select athletes at a later age (i.e., mid-late adolescence) with multiple entry and exit routes and coach- and athlete-nominations, demonstrating forward-thinking practices. The provision of high quality staff and holistic support was often limited by the resources available and a professional workforce of volunteers lacking formal qualifications. Players often experienced high training volumes and large travel commitments to access such talent development opportunities. These findings describe the structure and operationalisation of TIDS within global netball, highlighting common practices and potential shortfalls in delivering evidence-based talent identification and development practices. These findings should provide national and world netball organisations with factors to consider in improving their practices, which should focus upon coach and practitioner education and consider how organisations can collaborate to maximise the use of resources within the sport.

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ORCID

Fieke Rongen http://orcid.org/0000-0001-6367-8255

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